VIETNAM NATIONAL UNIVERSITY - HCM
Ho Chi Minh City University of Technology
Faculty of Computer Science and Engineering



MULTI-DISCIPLINARY PROJECT (CO3087)

Project

Self-Studying Management Systems

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Ho Chi Minh City, 28th July 2021

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DOCUMENT HISTORY

Date	Version	Changes	Person in charge
Apr 7	1.0	List out functional, non-functional requirements, devices needed	All
Apr 8	1.0.1	Write first version of report	All
Apr 13	1.0.2	Fix report and add use-case diagram	Trần Quốc Anh
Apr 16	1.0.3	Update functional requirements, add indices for use-case detail represented in use-case diagram	Nguyễn Hoàng Dũng
Apr 21	1.0.4	Update use-case detail	Nguyễn Hoàng Dũng
Apr 22	1.1	Add Mockup	All
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May 1	1.2.1	Add all Activity Diagrams (each use-case)	All
May 2	1.2.2	Add Deployment Diagram	All
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May 20	1.2.4	Update Evaluate real-time studying condition use-case	Đặng Trần Khánh
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Jun 24	1.5	Add Conclusion and Development Strategies	Nguyễn Hoàng Dũng
Jun 25	1.5.1	Fix content on Conclusion and Development Strategies	Trần Quốc Anh, Đặng Trần Khánh
Jul 4	1.5.2	Fix mock-up	All
Jul 26	1.6	Update class diagram	Nguyễn Hoàng Dũng

1 Introduction

Self-studying is a learning method where students direct their own studying outside the classroom and without direct supervision. Since students are able to take control of what (and how) they are learning, self-studying can be a very valuable way for many students to learn.

Also, since the knowledge nowadays develops exponentially, self-study is a must for every student. As a consequence, the amount of time required for students for self-studying is increasing.

However, the study space may affect the efficiency of self-studying. In Vietnam, most students take place for self-studying in the coffee shops, libraries,... instead of learning at home because of the distractions which appear in the house and students' bad self-management skill. The spread of the COVID-19 pandemic now make students unable to go to coffee shops and libraries, and almost 100% of their study time will be taken place in their study room.

With said in mind, we are working on a project to guide students to schedule their timetable well, as well as build a smart study space that can enhance the efficiency of self-studying.

In this project, we intend to use hardware devices for recording self-study's environmental conditions and notifying when it is time to study as he/she scheduled. An application is also provided to set schedule, check self-study performance and get advice to improve the study space.



Figure 1: Smart Study Space

2 Device

2.1 Input devices

1. ***Sound Sensor***
   - **Characteristics:** transform sound signal into electrical signal that the circuit can interpret. Size: 2.54mm x 2.54mm.
   - **Usage:** Check and validate the sound level, notify the users if the sound quality is unsuitable for learning problem.
2. ***Infrared sensor***
   - **Characteristics:** It can detect an obstacle effectively.
   - **Usage:** Check if the user is present at the study space(ex: desk), notify the user if he/she is not at the study space.
3. ***Real - time clock***
   - **Characteristics:** Record the real time value.
   - **Usage:** Assist the system in setting time, recording value.
4. ***DHT11***
   - **Characteristics:** Measure humidity(20% - 90% RH) and temperature(0 $\sim$ 50 °C).
   - **Usage:** Check if the temperature is comfortable for learning experience.
5. ***Light sensor***
   - **Characteristics:** Check the light condition.
   - **Usage:** Measure the light condition of the learning environment, notify the user if the study space is dark or dim(unsuitable for learning effectively).
6. ***Microbit***
   - **Characteristics:** This is the central control circuit.
   - **Usage:** Handle communication between server and devices.

2.2 Output devices

1. ***Buzzer***
   - **Characteristics:** Equipped with sound generating components that can be controlled by voltage input.
   - **Usage:** Notify the user of the event taking place(time for study, light condition too dark...).
2. ***LCD I2C***
   - **Characteristics:** It uses the I2C communication protocol and it is compatible with 3V3 and Microbit
   - **Usage:** Display the time and other notification.

3 User Requirements

3.1 Functional Requirements

- User can schedule (set the study time) the timetable in the application.
- The above schedule can be removed if the start time has not come yet.
- The application will be in charge of notifying when will be the study time or the break time. Normally, if the sessions takes more than 45 minutes, a 10-to-15-minute break time will be set up after 45 minutes of studying (to protect your spine).
- Data in the study space such as light level, noise,... will be recorded by sensors and transferred to the database. Notification is needed when an unsatisfying condition of light level, noise,... in the study space is met.
- The recorded data is also used to calculate study performance score of every self-study session, the score can be shown to user via the application with additional advice.
- Hardware devices will detect whether student appears in the study session as set previously in the application or not. Mobile application will send notification if user is not in the study place at that time.

3.2 Non-Functional Requi̇rements

- Easy to use.
- Data transfers and queries between user mobile application and device must be at maximum 2 seconds for each actions.
- The system must ensure working correctly 24/7.
- Application can support multi-platform.
- The application is compatible in both iOS and Android operating system.

4 Use-case Detail

All Implementation views (i.e., Activity diagrams) appear after each use-case detail can be accessed here in case they are too difficult to observe.

4.1 Manage study schedule

Use Case ID:	1		
Use Case Name:	Schedule new study sessions		
Create By:	Nguyễn Khắc Hào	**Last Updated By:**	Nguyễn Khắc Hào
Date Created:	8/4/2021	**Date Last Updated:**	23/6/2021
Actors:	User		
Description:	Set or modify the study schedule		
Preconditions:	The user has logged in the application		
Normal Flow:	1. The user opens the application and selects "Show my schedule" from the home screen or "Schedule" from the side menu. 2. The user selects "Add Event". 3. The application displays a form requesting information required to create a new study event. 4. The user fills in the form. 5. The user selects "Save Event". 6. The application checks the entered information's validity and sends an add event request to the server. 7. The application returns to the Schedule screen and displays an updated schedule.		
Alternative Flow:	Alternative 1 in step 5: The user selects the return button and returns to the Schedule screen without adding the new event. Alternative 1 in step 6: If the user haven't entered any field, the application will not send a request to the server and show an error dialog, return to step 4. Alternative 2 in step 6: If the event has an end time that is sooner than its start time, the application will not send a request to the server and show an error dialog, return to step 4. Alternative 3 in step 6: If there exists an event that overlap the current request in terms of start time and end time, the application will not send a request to the server and show an error dialog, return to step 4.		
Exceptions:	Exception in step 6: If fail to connect to the server for pushing new schedule, the application will show an error dialog and return user to step 1.		

Use Case ID:	2		
Use Case Name:	Review schedule		
Create By:	Nguyễn Khắc Hào	**Last Updated By:**	Nguyễn Khắc Hào
Date Created:	8/4/2021	**Date Last Updated:**	23/6/2021
Actors:	User		
Description:	View upcoming study events		
Preconditions:	The user has logged in the application		
Normal Flow:	1. The user opens the application and selects "Show my schedule" from the home screen or "Schedule" from the side menu. 2. The application requests and retrieves from the server all upcoming study events within a time frame. 3. The application display a schedule (timeline) view, along with a list of upcoming schedule within a time frame. 4. The user selects a study event in the list to see more detailed information about that event. 5. The application displays more information about the selected event.		
Alternative Flow:	Alternative 1 in step 4: The user can change the schedule's time frame ("Today", "This Week", "This Month", ...). The schedule will change its display mode accordingly.		
Exceptions:	Exception 1 in step 3: If the application fails to retrieve data from the server, an error message will be displayed instead, user return to step 1.		

Use Case ID:	3		
Use Case Name:	Remove study sessions		
Create By:	Nguyễn Khắc Hào	**Last Updated By:**	Nguyễn Khắc Hào
Date Created:	8/6/2021	**Date Last Updated:**	23/6/2021
Actors:	User		
Description:	Remove upcoming study events from the schedule		
Preconditions:	The user is logged in the application and has at least one study event scheduled in the future.		
Normal Flow:	1. The user opens the application and selects "Show my schedule" from the home screen or "Schedule" from the side menu. 2. The application requests and retrieves from the server all upcoming study events within a time frame. 3. The application display a schedule (timeline) view, along with a list of upcoming schedule within a time frame. 4. The user selects a study event in the list to see more detailed information about that event. 5. The application displays more information about the selected event, with an option to remove that event. 6. The user selects "Remove this event" option. 7. The application sends a request to the server to remove the event from the database. 8. The application returns to Schedule screen and displays an updated schedule.		
Alternative Flow:	Alternative 1 in step 4: The user selects the return button and returns to the Schedule screen without removing the event.		
Exceptions:	Exception 1 in step 8: If the application fails to communicate with the server, it will return to the Schedule screen and display the previously loaded schedule instead.		

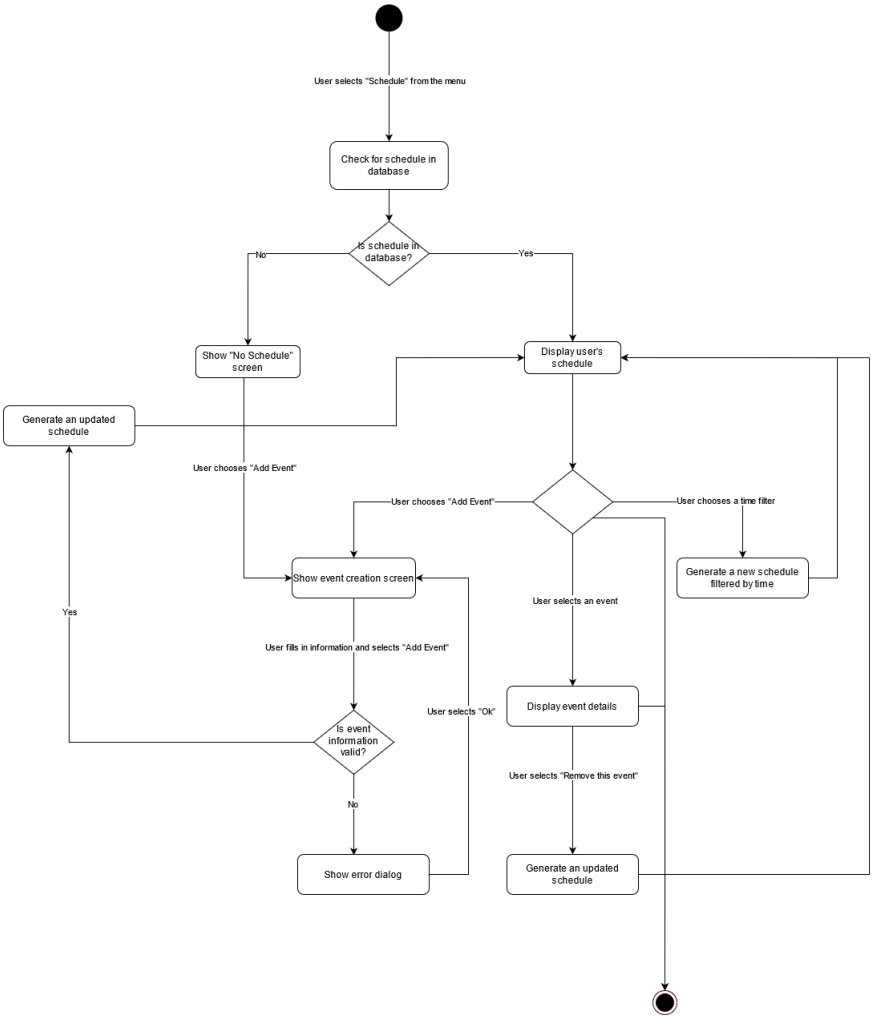


Figure 2: *Activity diagram of Manage study schedule*

4.2 Notify study time/ break time

Use Case ID:	4		
Use Case Name:	Notify at scheduled session time		
Create By:	Trần Quốc Anh	**Last Updated By:**	Trần Quốc Anh
Date Created:	8/4/2021	**Date Last Updated:**	23/6/2021
Actors:	User		
Description:	Send notification with a given study session and when the break time begin/end.		
Preconditions:	User has logged in the application.		
Normal Flow:	1. The devices get data about study session time and end time from database. 2. The application will set notification based on scheduled time received. 3. At the start time of study session, notifications will appear with sound and vibration. 4. At the end time of the study session, notifications will also appear.		
Alternative Flow:	Alternative 1 in step 3: If presence switch in setting is turned on, after start time of each session 5 minutes, the application will get latest data from server to check the presence of user, if the user is not sitting in front of table, a notification will appear, else nothing happen. Alternative 2 in step 3: If break switch in settings is turned on, after start time of each session 45 minutes (in the final demo it will be modified to 30 seconds), a notification will appear to remind about break time.		

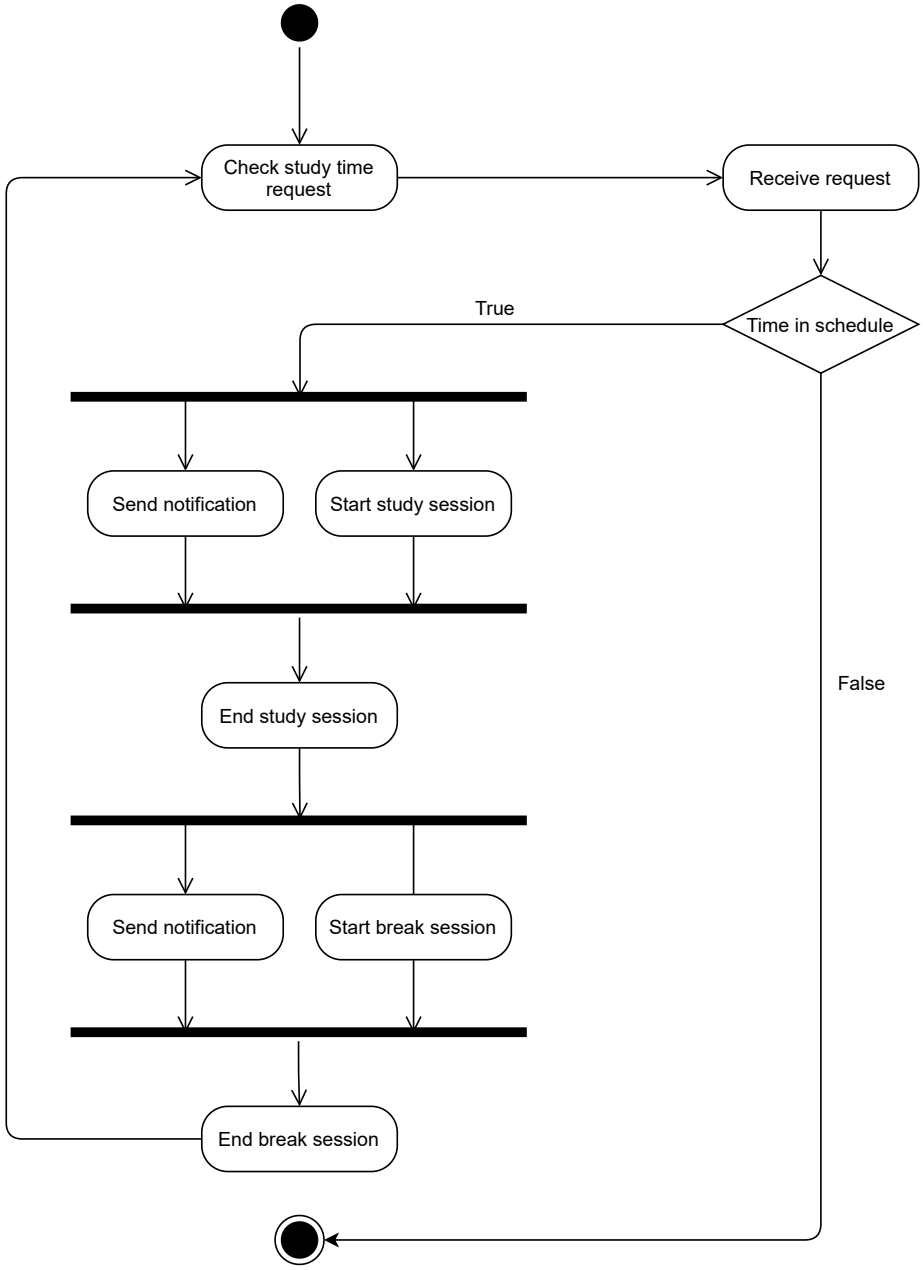


Figure 3: *Activity diagram of Notify at scheduled session time*

4.3 Detect student's presence

Use Case ID:	5		
Use Case Name:	Detect student's presence		
Create By:	Nguyễn Lâm Nguyên Bảo	**Last Updated By:**	Nguyễn Lâm Nguyên Bảo
Date Created:	8/4/2021	**Date Last Updated:**	20/6/2021
Actors:	Infrared Sensor (Hardware Device)		
Description:	Check if student is at the study place at specified time		
Preconditions:	1. User has already set up the timetable on the device. 2. The setting for this function is enabled.		
Normal Flow:	1. The system periodically checks for upcoming study sessions. 2. If there is a session about to start, the app will establish a connection to the infrared sensor. 3. The infrared sensors checks if the student is present at the study space by the time the session starts. 4. If the sensors return 0, meaning the student was not detected, the app will send a notification. 5. When the student is detected, no notification will be sent and the system continues the normal operation flow.		
Exceptions:	Exception in step 3: - If fail to establish a connect to the sensor after a certain number of times, the flow cannot continues.		

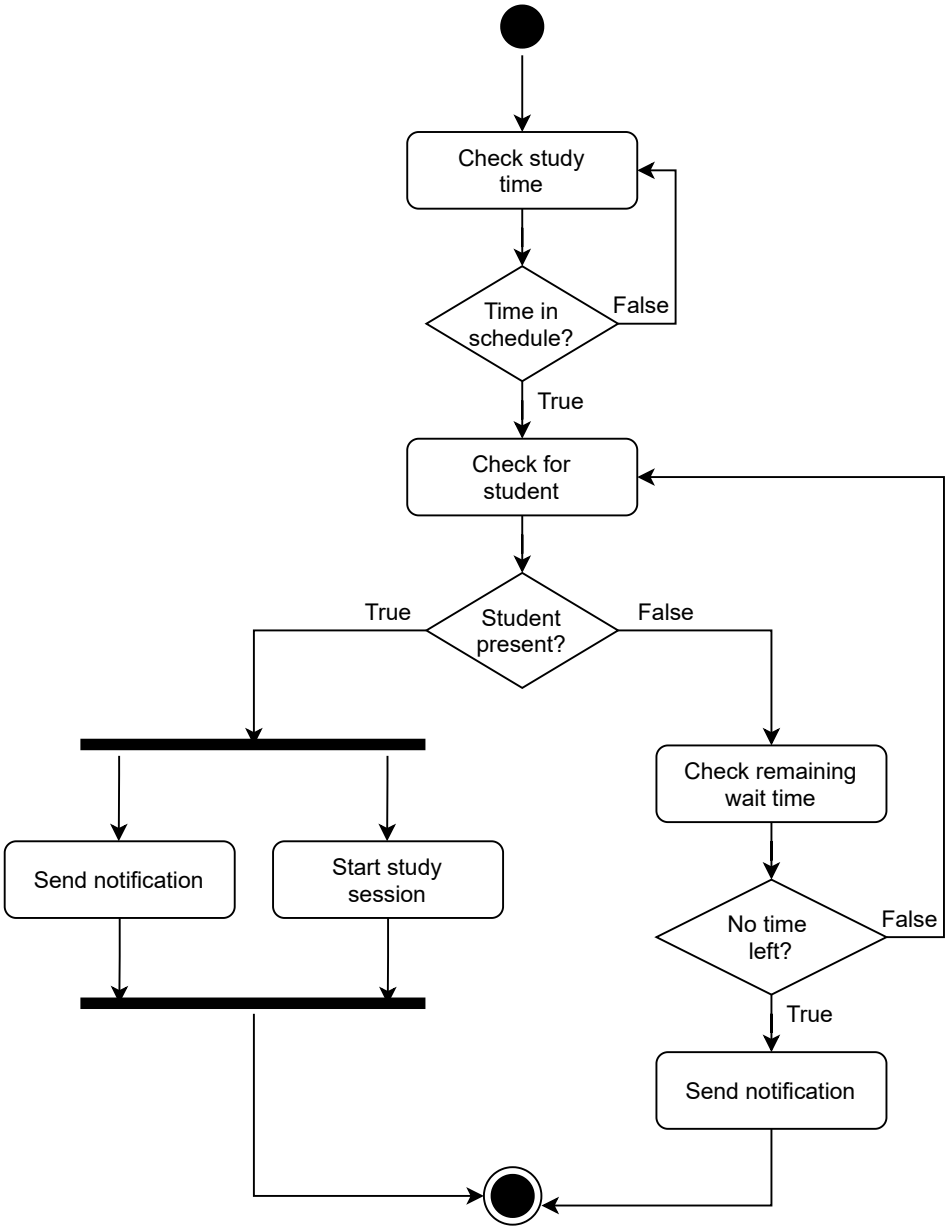


Figure 4: *Activity diagram of Detect student's presence*

4.4 Evaluate real-time studying condition

Use Case ID:	6		
Use Case Name:	Evaluate real-time studying condition		
Create By:	Đặng Trần Khánh	**Last Updated By:**	Đặng Trần Khánh
Date Created:	5/6/2021	**Date Last Updated:**	23/4/2021
User story:	As a student preparing for the exam, I want the most ideal learning environment that minimizes the chance of encountering distracting factors such as noise so I can learn more productively		
Actors:	Hardware devices		
Description:	Evaluate the study environment by collecting the data measured by the sensors, give warning accordingly if the condition is not suitable for the best experience.		
Preconditions:	Users are using the application and is verified as learning in the study space (using infrared sensor to detect objects). All the sensors are functional.		
Normal Flow:	1. The devices(light sensor, sound sensor, heat sensor) collect the data from the environment. 2. The data are sent to the server and processed, compared to benchmark 3. Users opens section sensor in the menu to access the available sensors being used. 4. Users can select any of the sensors to view detailed data and manually adjust the settings according to their preference. 5. If the data violate the condition, the application sends notification to output devices 6. Output devices (LCD, buzzer) receive signal from the application and notify the user		
Alternative Flow:	Alternative 1 in step 4: - Users can open the predefined list of values and their indications, they can select value according to this list. Alternative 2 in step 4: - Users can choose to switch off any sensors. Alternative 1 in step 6: -If the data are within allowed range, the system will stay idle.		
Exceptions	Exception in step 2: - The sensors fail to connect to the application and send the data, connection warning will be raised. - The result must be displayed in real time, raise warning if the execution time exceeds 1 second.		

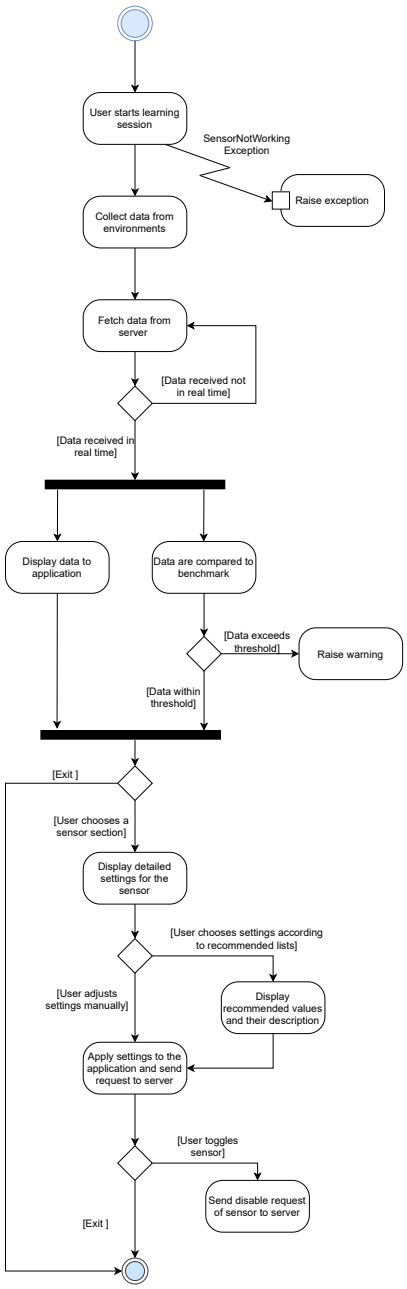


Figure 5: *Activity diagram of Evaluate real-time studying condition*

4.5 Show study performance

Use Case ID:	7		
Use Case Name:	Show study performance		
Create By:	Nguyễn Hoàng Dũng	**Last Updated By:**	Nguyễn Hoàng Dũng
Date Created:	8/4/2021	**Date Last Updated:**	8/6/2021
Actors:	Users		
Description:	Display the study status in one/all study session(s).		
Preconditions:	User has logged in the application.		
Normal Flow:	1. User opens the application and select "Summary". 2. The application shows the study score of last study sessions. 3. User choose a displayed study session to see the full detail. 4. The application shows the detail. 5. User chooses 1 factor (light, noise, temperature) to know the full detail. 6. The application shows the status of the factor in the specific study sessions (graph and advice). 7. User exit the 'Session' screen or navigate to other screen.		
Alternative Flow:	Alternative 1 in step 2: - If there is a recently studied session that have not had the performance score, the system will evaluate the score from recorded sensor data and store it in the database. Alternative 1 in step 3: - User can refresh the session screen if there is a newly updated session. Alternative 2 in step 3: - User can modify number of displaying study session on the screen. Alternative 3 in step 3: - User can sort the displaying study session by categories (performance score, session name, time, number of bad conditions). Alternative 1 in step 7: - User can tap on the Question button to see why the system gave that performance score.		
Exceptions:	Exception in step 2: - When there is no session set by the user completed, move to step 7. Exception in step 4: - When there is no recorded sensor data, the graph and statistic results will be unavailable, move to step 7.		

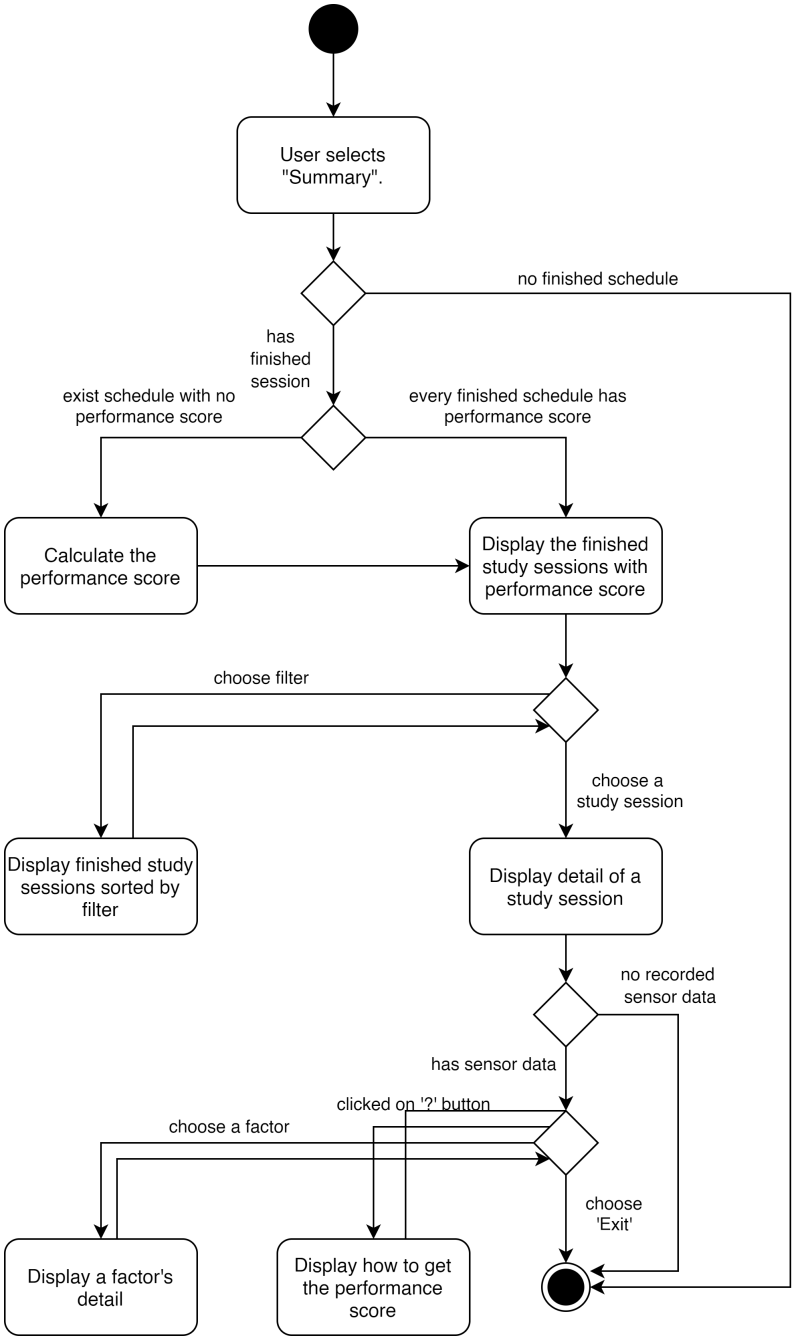


Figure 6: *Activity diagram of Show study performance*

5 Use-case Diagram

There are 2 use-cases (*Show current sensor data* and *Login/Logout/Signup*) that are not mentioned in the use-case detail because of its simplicity.

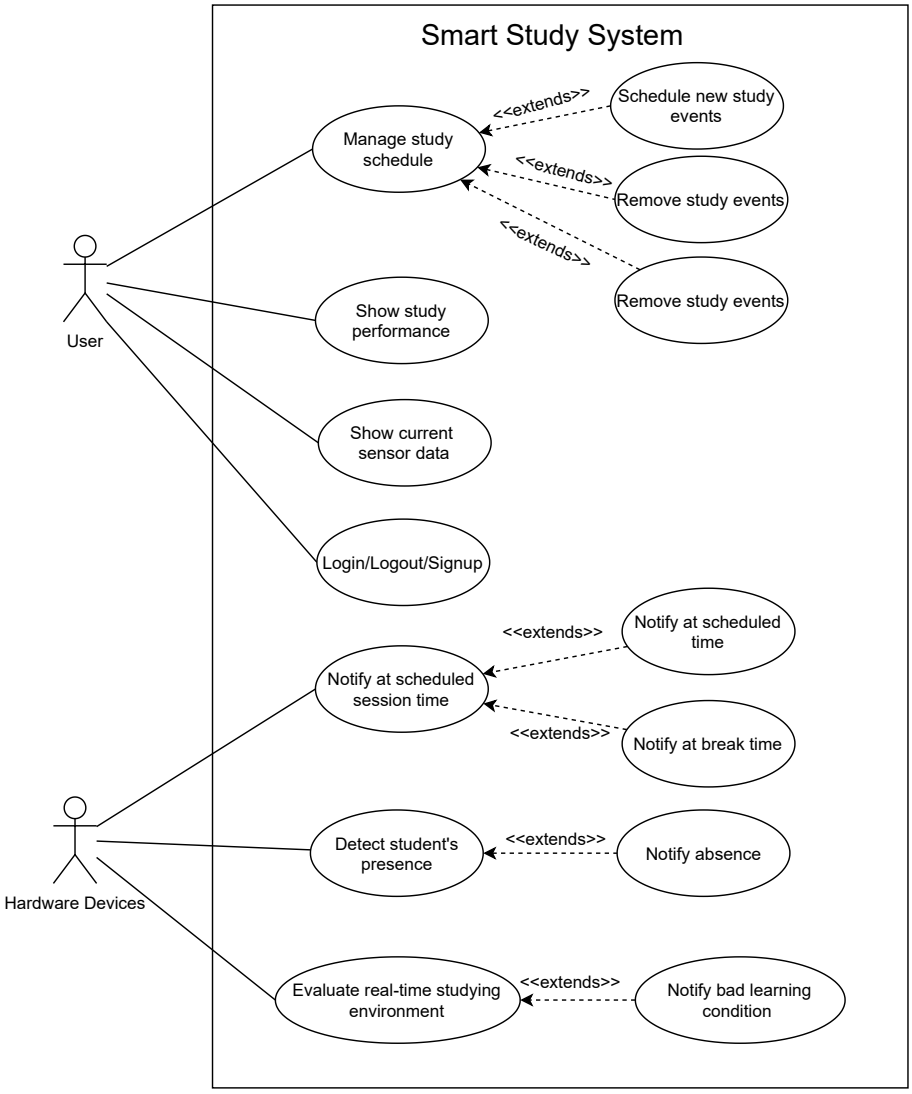


Figure 7: *Use case diagram*

6 Mockup

6.1 Welcome, authentication page

The images below are the welcome page when users first open the app, the sign up and log in pages for authentication.

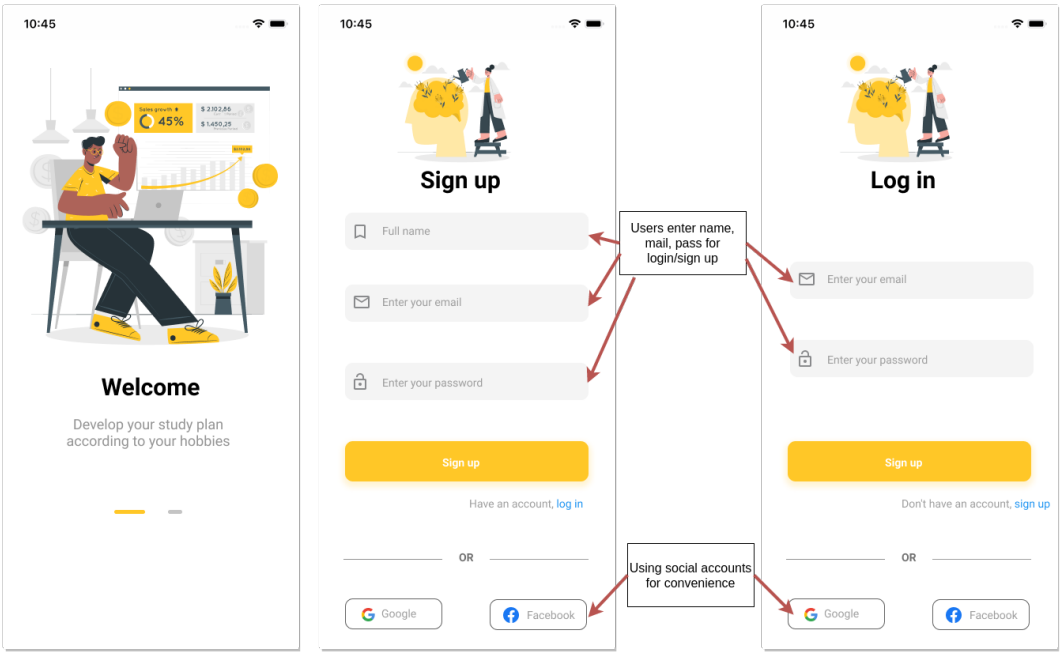


Figure 8: *Welcome, Sign up and Log in page*

6.2 Homepage and Navigation

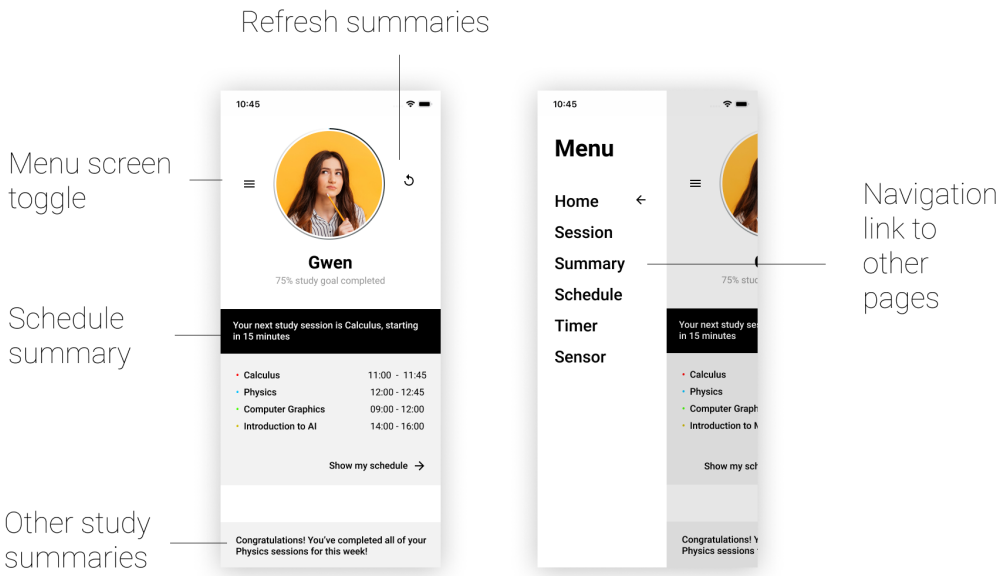


Figure 9: *Home Screen and Menu Screen.*

6.3 Notify at scheduled session time

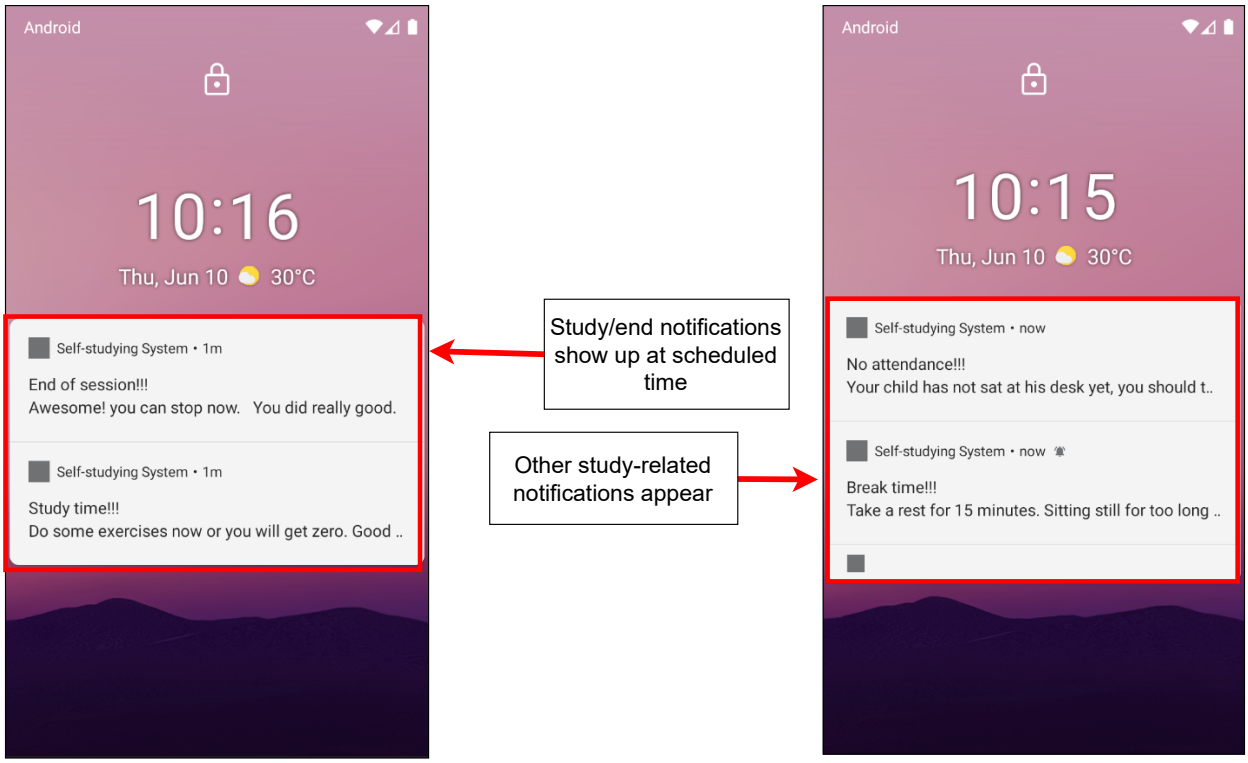


Figure 10: *Notifications show up at scheduled time.*

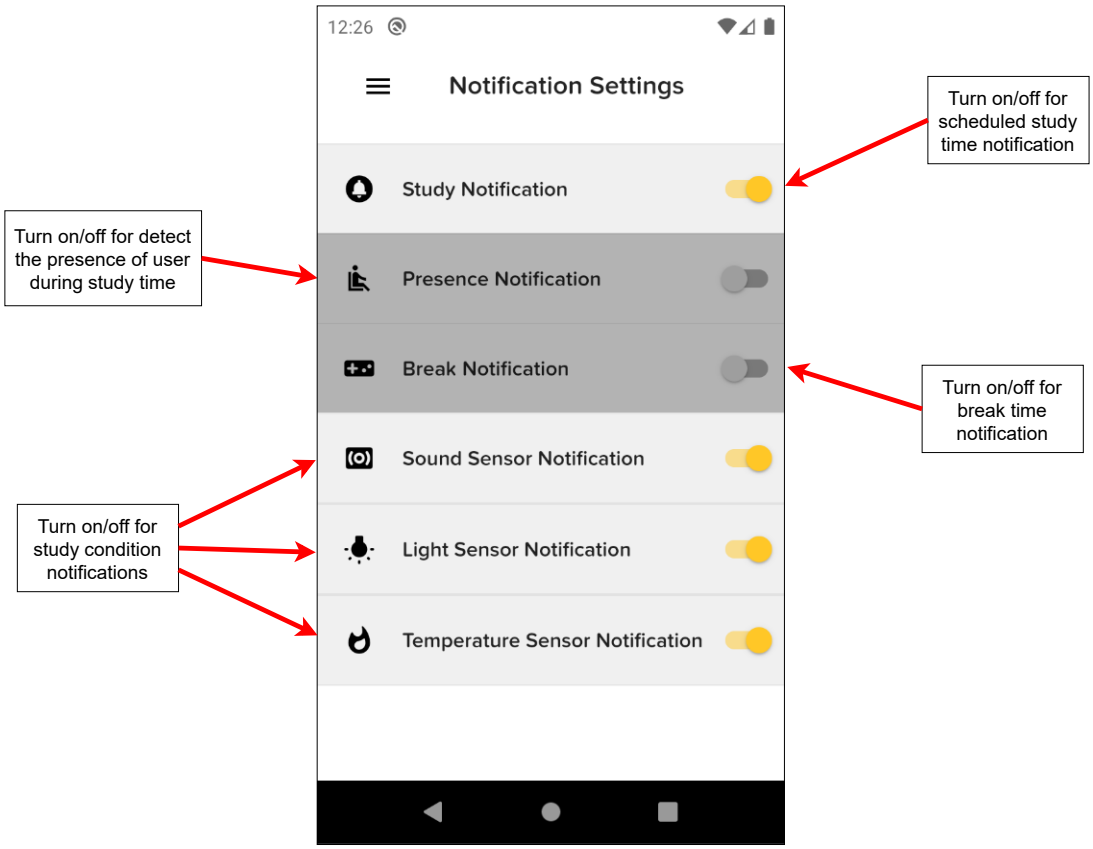


Figure 11: *Notification settings screen*

6.4 Show study performance

- While the application is receiving data, the list view will display a text *'Loading...'* with a note in case there is a sessions that have just finished and have not processed its score, it would take a bit more time to load the schedules.
- The same cases for *"Loading..."* text in detailed schedule screen while loading sensor data.
- The color of score circle may have either 4 types: green (when score $>$ 90, updated from the mock up where needs score $=$ 100), yellow (90 $>$ score $>$ 70), red (70 $>$ score $>$ 0), grey (score $=$ 0 or no recorded data received).

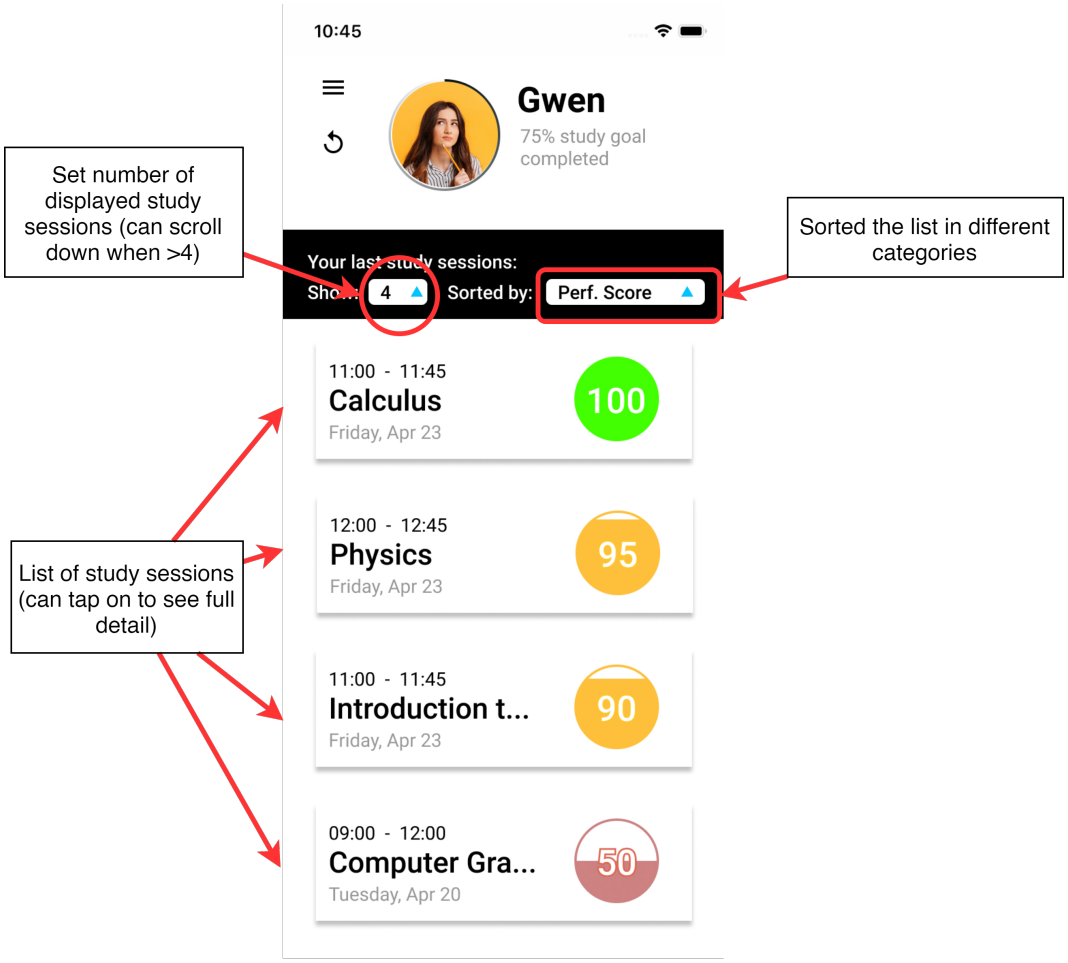


Figure 12: *Main screen of study summary for showing study performance*

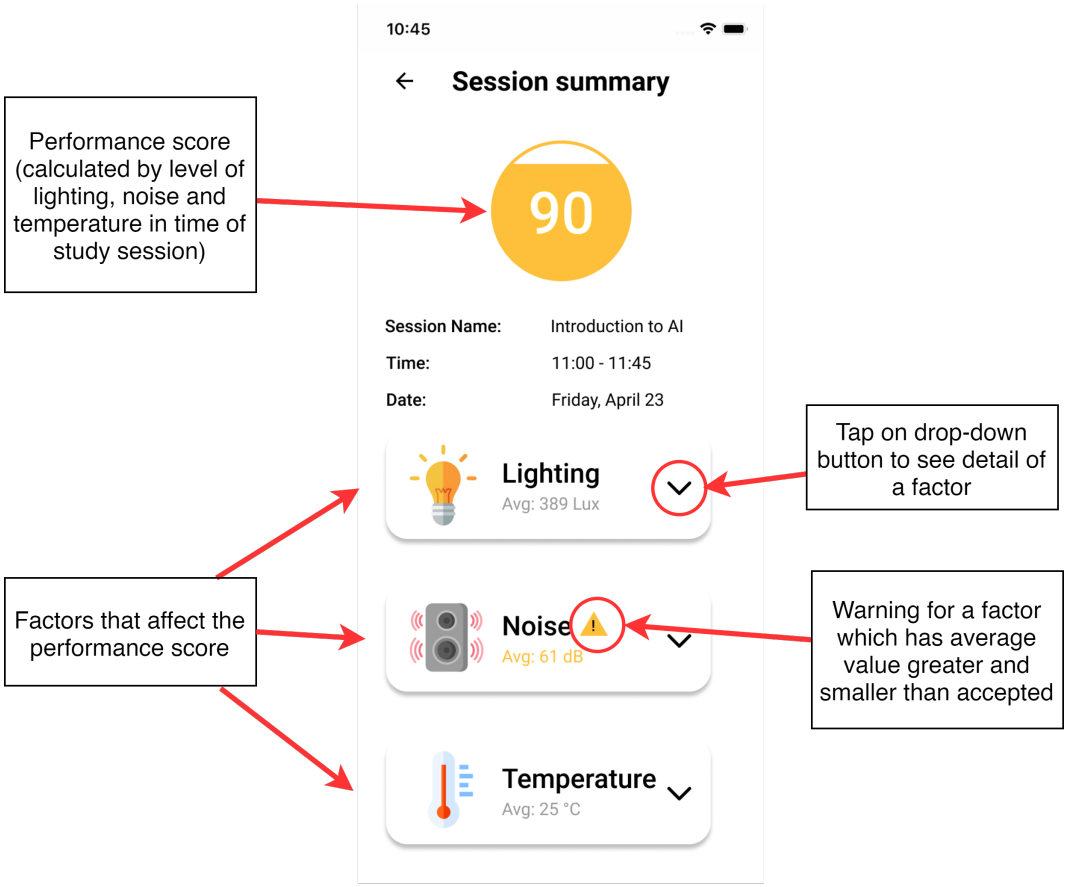


Figure 13: *One study session summary*

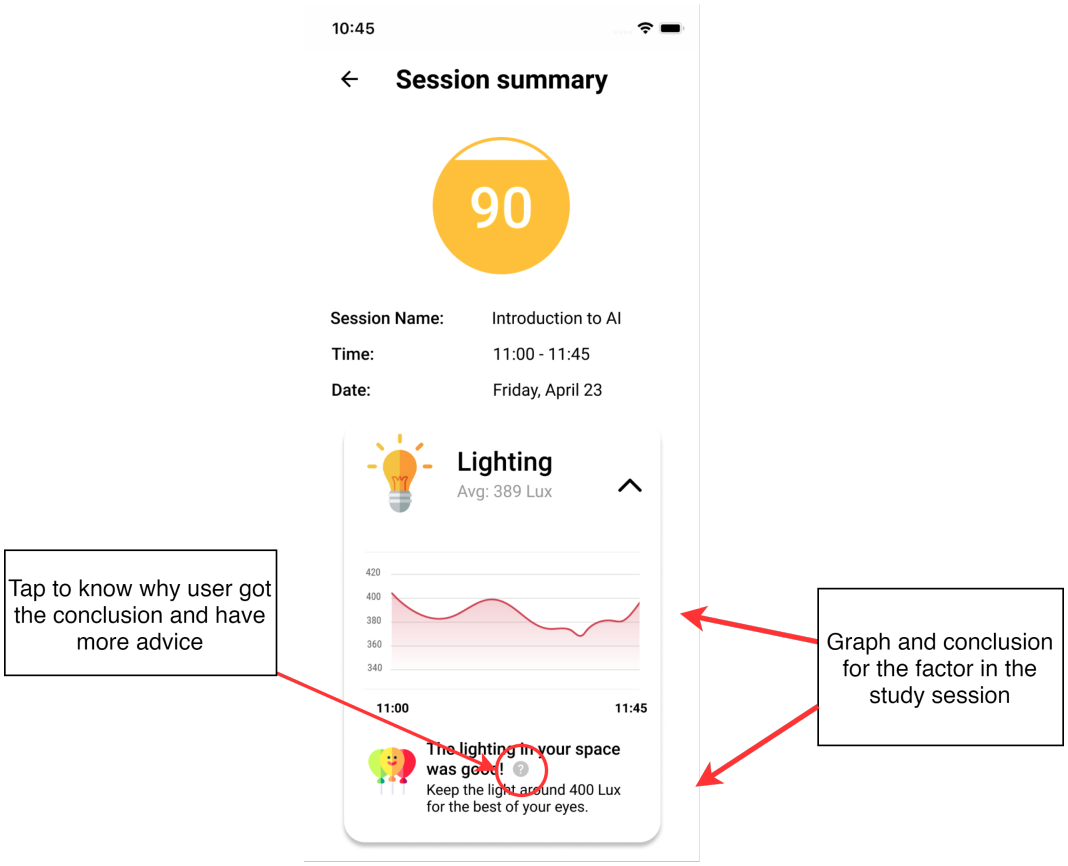


Figure 14: *Detail in a session*

6.5 Schedule management

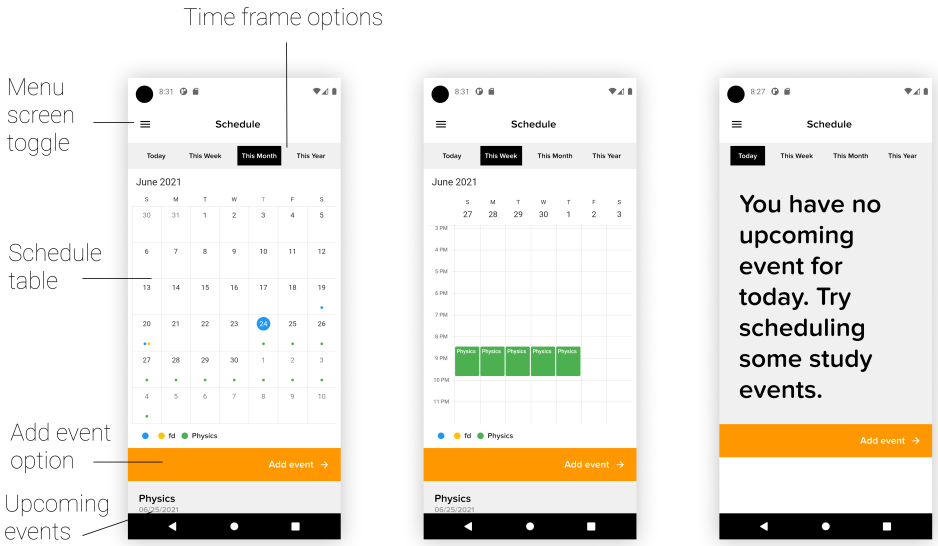


Figure 15: *Schedule review screens*

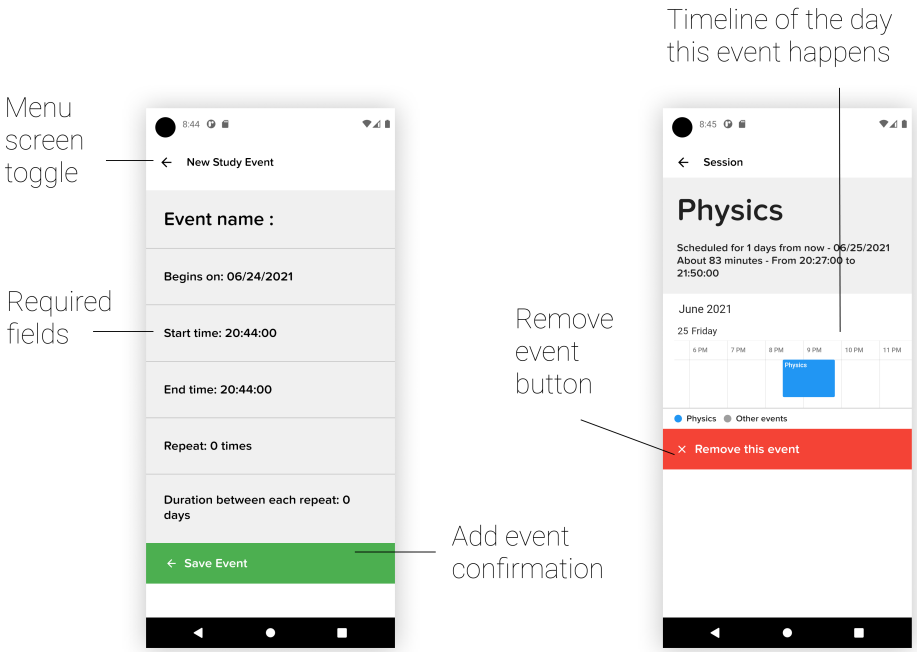


Figure 16: *Schedule's Add Event and Event Detail screens*

6.6 Evaluate real-time studying condition

Regarding studying condition, we have 3 different sensors for 3 tasks:

- Lighting sensor to give user the measure of intensity of light within the study area, raise warning if it's too dark.
- Sound sensor to measure the sound level, raise warning if the environment exceeds acceptable noise threshold.
- Temperature to measure the room temperature, raise warning if the room temperature may yield uncomfortable learning experience.

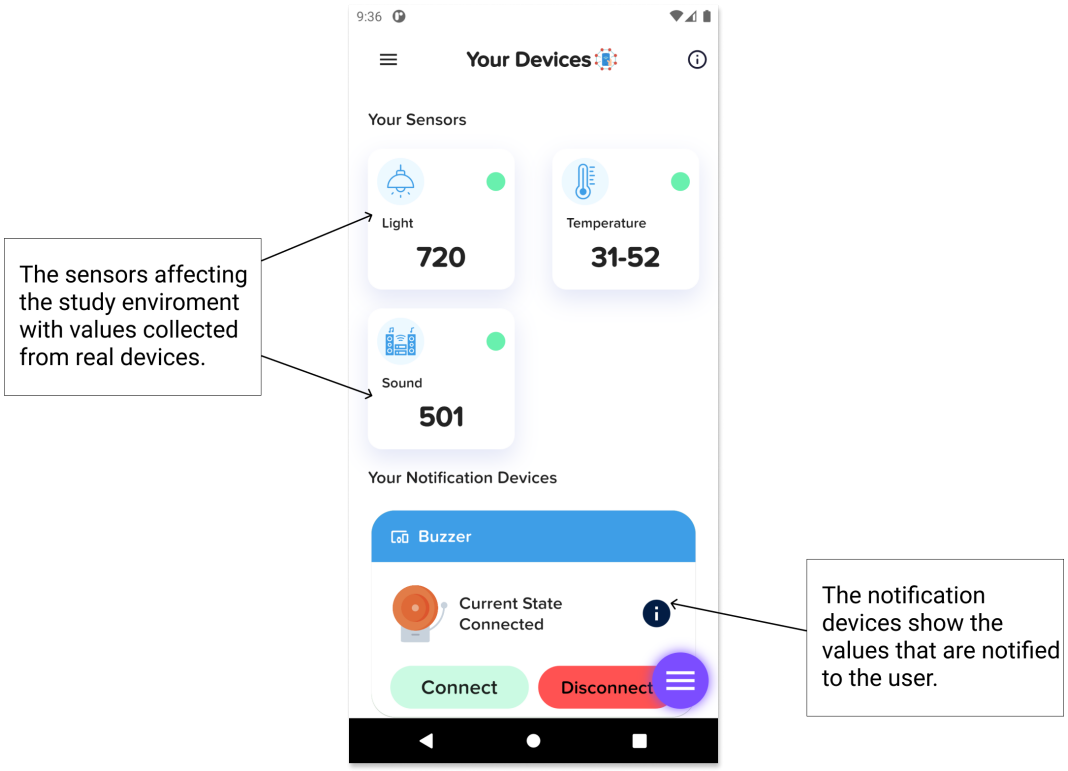


Figure 17: *The three sensors for measuring study environments*

Tapping on one of the available sensors will lead users to this screen where they can toggle the sensors if they do not wish to be disturbed by it, manually set the range to their preference or view the list of values and their indications (for example: 130db to indicate the sound level surpass the threshold of pain -> unsuitable for learning)

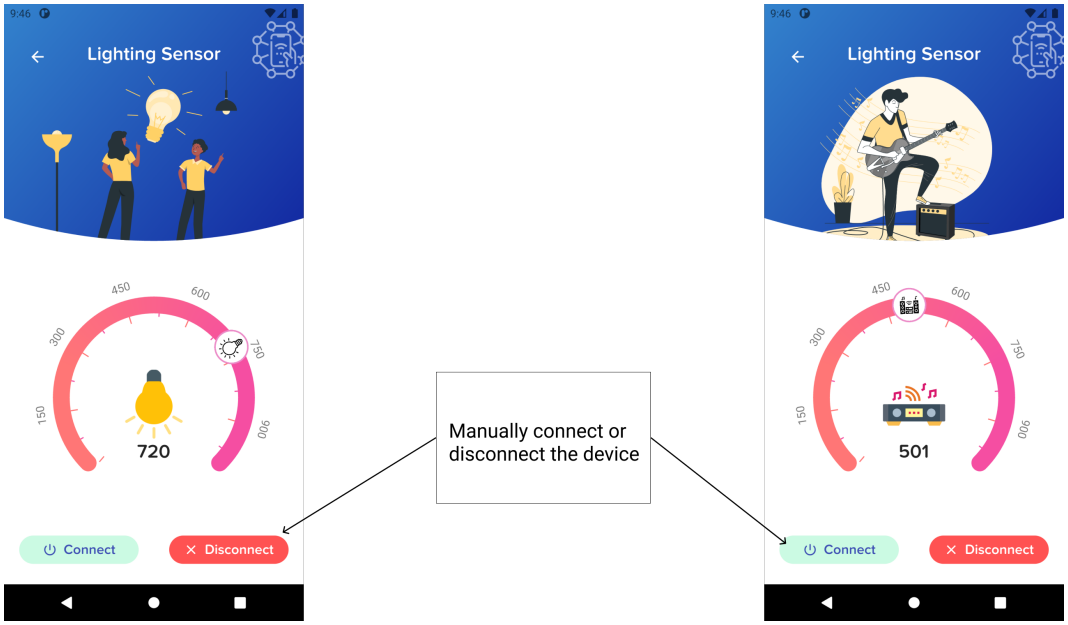


Figure 18: *Settings for each sensor*

7 Class diagram

7.1 Clarification

The figure of class diagram is shown in Figure 19. However, it may be difficult to observe in this document, readers should get on this link for .drawio file or this link for .jpg file.

The whole project is divided into 3+1 blocks as can be seen from the class diagram.

- The **Controllers** block, whose classes shown with blue headers (except for SensorEvaluation as a enumeration), is responsible for taking the requests from view block and do the appropriate operations. Most of the business logic in defined here, especially the logic to get query/information from the servers (e.g., when the user asks for showing all the upcoming session, it will call the SessionController to handle this. The SessionController will establish a connection to SQLServer to do a query, then fetch the result into JSON format, the result will be convert into a Session object that will need to display on the view).
- The **Model** block, whose classes shown with red headers, stored necessary information and generate suitable format for showing on the views. It can only be created by classes in the Controller block.
- The **View** block, whose classes shown with yellow headers, and the block is divided into smaller blocks to define members' work as well as get more convenience while walking through the use-case. It provides the interface for user and call the appropriate controllers every time user has a request.
- The 2 classes with purple headers are **servers**, these are not implemented in the application itself but in external web servers.

The actual implementation may appear minor difference from this to suit certain desires and frameworks.

7.2 Class diagram



Figure 19: *Class diagram*

8 Deployment View

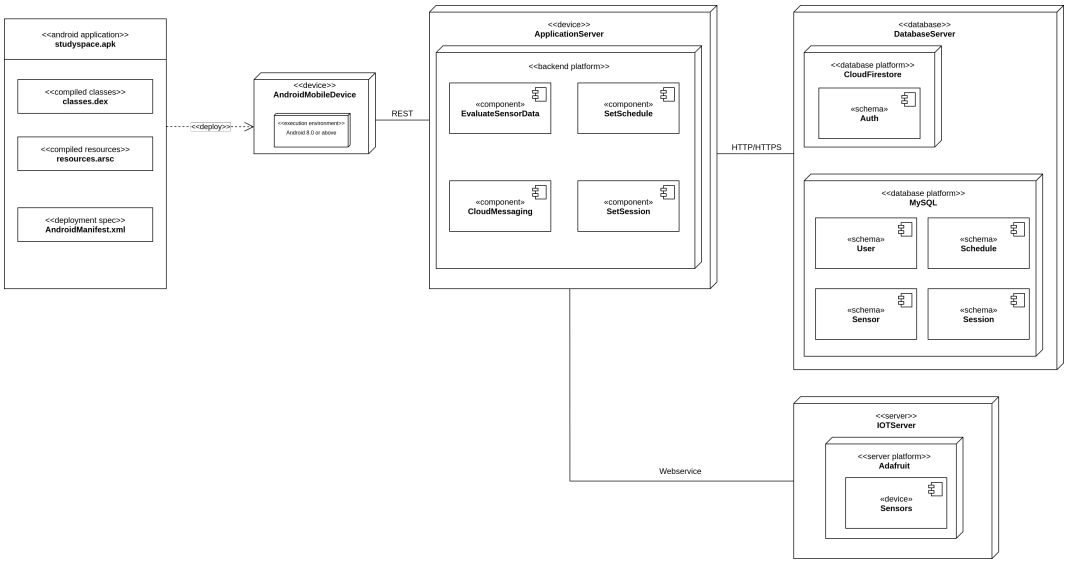


Figure 20: *Deployment diagram*

9 Database Design

9.1 Entity-Relationship Diagram (ER diagram)

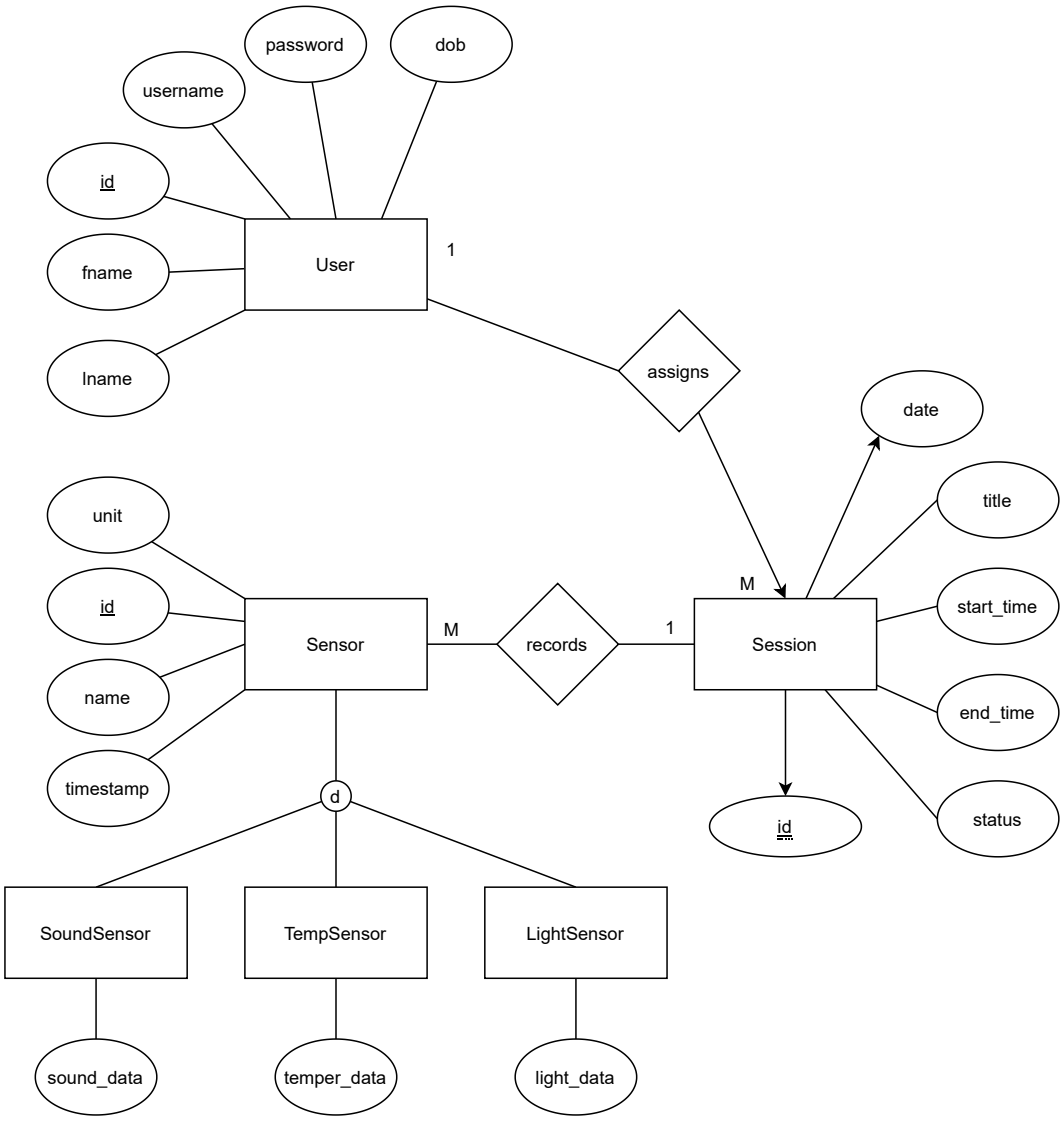


Figure 21: *ER diagram*

9.2 Relation Schema

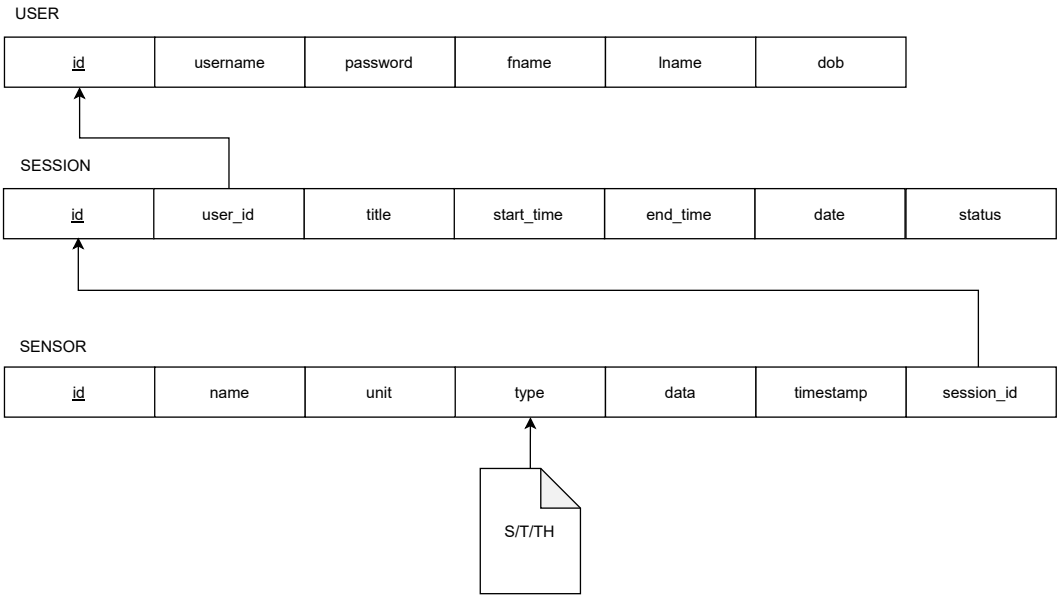


Figure 22: *Relation schema*

10 Conclusion and Development Strategies

10.1 Summary

Here are the things that we have achieved in this project:

- Successfully organizing an application to be the connection between hardware devices (i.e., sensors) and user for the aim to raise awareness of improving environmental conditions around user's study space to study more efficient.
- Walking through the process of Software Engineering once again after the project from *Practice on Software Engineering* last year.
- Applying the knowledge of Database System to manage business data.
- Gaining skills of frameworks: Flutter, MySQL, Firebase, MQTT, Flask (Python), etc.
- Teamwork skills.

10.2 Self-evaluation

The team with 5 members have made great contributions to the project. Accordingly,

- **Trần Quốc Anh**: Work with MQTT server, develop the use-case of notifying study start/end time, break time and detect student's presence, organize a web server as a service.
- **Nguyễn Lâm Nguyên Bảo**: Improve the relation schema, develop the MySQL server, work with controllers.
- **Nguyễn Hoàng Dũng**: Improve UI/UX, develop the use-case of show study performance, take some work at servers (MySQL, MQTT/AdaFruit), improve class diagram, complete the report and related documents.
- **Nguyễn Khắc Hào**: Develop the use-case of managing study schedule, improve UI, improve the relation schema.
- **Đặng Trần Khánh**: Develop the use-case of evaluate real-time studying condition, work most with MQTT/Adafruit server, organize a web server as a service, improve deployment view.

10.3 Development Strategies

From the thought that we have not fully developed, here are the potential suggestions for development in the future whenever there is a chance to continue the project:

- Provide a platform to consult user the best equipment to buy for their study space from their room specification and user's preferred style. If success, the application will be a connection for user and equipment's seller and be considered as in commercial use.
- The application will separate user roles to be *Student Mode* and *Parent Mode*. The notifications will be raised for appropriate user in a particular mode. Parent can observe the study status of their child.
- Integrate the quiz system that allows users to test themselves when the need arises with numerous tests from different subjects.
- Suggest study space design according to users' preference, personalities to further enhance the learning experience.